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TOPIC: APPLICATION OF STACK

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SUBMITTED TO: MA'AM NEHAN

QUESTION: 1

Convert Infix $\rightarrow$ Postfix

Infix:

$$1. A+B * (C^D - E)^{(F+G*H)} - I$$

Input Expression:	Stack: (bottom $\rightarrow$ top)	Postfix Expression:
A		A
+	+	A
B	+	AB
*	+*	AB
(	+*(	AB
C	+*(	ABC
^	+*(^	ABC
D	+*(^	ABCD
-	+*(-	ABCD^
E	+*(-	ABCD^E
)	+*(-)	ABCD^E
	+*	ABCD^E-
^	+*^	ABCD^E-
(	+*^(	ABCD^E-
F	+*^(	ABCD^E-F
+	+*^(+	ABCD^E-F
G	+*^(+	ABCD^E-FG
*	+*^(+*	ABCD^E-FG
H	+*^(+*	ABCD^E-FGH
)	+*^(+*)	ABCD^E-FGH
	+*^	ABCD^E-FGH*+

-	+ -	$ABCD^E - FGH^* + ^* I$
I	+ -	$ABCD^E - FGH^* + ^* I$
End	Popping stack	$ABCD^E - FGH^* + ^* I - +$

Final Postfix Expression:

$ABCD^E - FGH^* + ^* I - +$

2. $A + (B * C - (D / E ^ F) * G) * H$

Input Expression:	Stack: (bottom → top)	Postfix Expression:
A		A
+	+	A
(	+(	A
B	+(	AB
*	+(*	AB
C	+(*	ABC
-	+(-	ABC*
(	+(- (	ABC*
D	+(- (	ABC*D
/	+(- (/	ABC*D
E	+(- (/	ABC*DE
^	+(- (/ ^	ABC*DE
F	+(- (/ ^	ABC*DEF

)	+ (- (/ ^)	ABC * DEF
	+ (-	ABC * DEF ^ /
*	+ (- *	ABC * DEF ^ /
G	+ (- *	ABC * DEF ^ / G
)	+ (- *	ABC * DEF ^ / G
	+	ABC * DEF ^ / G * -
*	+	ABC * DEF ^ / G * -
H	+	ABC * DEF ^ / G * - H
End	Popping stack	ABC * DEF ^ / G * - H * +

Final Postfix Expression:

$ABC * DEF ^ / G * - H * +$

QUESTION: 2

Convert Infix $\longrightarrow$ Prefix:

i) $A + (B * (C + (D * E)))$

Reverse the Expression:

$))) E * D (+ C (* B (+ A$

Swapping Parentheses:

$(((E * D) + C) * B) + A$

Symbol:	Stack:	Output:
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(	(	
---	---	--

(	((	
---	----	--

(	((	E
E	((	E
*	((*	ED
D	((*	ED
)	((*)	ED*
	((	ED*
+	((+	ED*C
C	((+	ED*C
)	((+)	ED*C+
	(	ED*C+
*	(*	ED*C+B
B	(*	ED*C+B
)	(*)	ED*C+B*
		ED*C+B*
+	+	ED*C+B*A
A	+	ED*C+B*A+
End	Popping stack	
	Reversing Output:	+ A*B+C*DE

Final Prefix Expression:

+A*B+C*DE

2. $A * (B + C * (D - E)) + F / G$

Reversing Expression:

$G / F +)) E - D (* C + B (* A$

Swapping Parentheses:

$G / F + ((E - D) * C + B) * A$

Symbol:	Stack:	Output:
G		G
/	/	G
F	/	GF
+	+	GF/
(	+(	GF/
(	+((	GF/
E	+((	GF/E
-	+((-	GF/E
D	+((-	GF/ED
)	+((-)	GF/ED
	+(	GF/ED -
*	+(*	GF/ED -
C	+(*	GF/ED - C
+	+(+	GF/ED - C *
B	+(+	GF/ED - C * B
)	+(+)	GF/ED - C * B
	+	GF/ED - C * B +
*	+ *	GF/ED - C * B +
A	+ *	GF/ED - C * B + A
End	Popping stack	GF/ED - C * B + A * +
	Reversed Output:	+ * A + B * C - DE / FG

Final Prefix Expression:

$$+ * A + B * C - DE / FG$$

QUESTION: 3

Evaluate the following Expressions:

• $56 + 2 * 34 + -$

1. Postfix:

Symbol:	Stack:	Operation:
5	5	
6	5, 6	
+	11	$5 + 6 = 11$
2	11, 2	
*	22	$11 * 2 = 22$
3	22, 3	
4	22, 3, 4	
+	22, 7	$3 + 4 = 7$
-	15	$22 - 7 = 15$

Result: 15

• $- * 45 + 23$

2. Prefix:

Scanning right to left

P.T.O

Symbol:	Stack:	Operation:
3	3	
2	3, 2	
+	5	$2+3=5$
5	5, 5	
4	5, 5, 4	
*	5, 20	$4*5=20$
-	15	$20-5=15$

Result = 15

• + * 2 3 4

3. Prefix

Scan right to left.

Symbol:	Stack	Operation:
4	4	
3	4, 3	
2	4, 3, 2	
*	4, 6	$2*3$
+	10	$6+4=10$

Result: 10

• $93 * 42 / -$

4. Postfix:

Scan left to right

Symbol:	Stack:	Operation:
9	9	
3	9, 3	
*	27	$3 * 9$
4	27, 4	
2	27, 4, 2	
/	27, 2	$4 / 2 = 2$
-	25	$27 - 2 = 25$

Result: 25
